Hallmarks Cell Stem Cell

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Review

Metabolic regulation of the hallmarks of stem cell biology

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SUMMARY

Stem cells perform many different functions, each of which requires specific metabolic adaptations. Over the past decades, studies of pluripotent and tissue stem cells have uncovered a range of metabolic preferences and strategies that correlate with or exert control over specific cell states. This review aims to describe the common themes that emerge from the study of stem cell metabolism: (1) metabolic pathways supporting stem cell proliferation, (2) metabolic pathways maintaining stem cell quiescence, (3) metabolic control of cellular stress responses and cell death, (4) metabolic regulation of stem cell identity, and (5) metabolic requirements of the stem cell niche.

INTRODUCTION

Stem cells have two jobs: to self-renew and to differentiate. The ability of stem cells to expand and give rise to specialized progeny underlies development, tissue regeneration, and normal homeostatic organismal function. Disrupting the balance between self-renewal and differentiation results in a wide range of diseases from developmental disorders to cancer. Accordingly, stem cell quiescence, proliferation, and differentiation are carefully guarded by factors including niche-derived signals, signaling cascades, and transcriptional networks.

Cell metabolism is emerging as an additional determinant of stem cell function. Cellular metabolic networks are not static, but rather can be tuned to meet metabolic demands of specific cell types.[1] Proliferating cells that require continuous duplication of biomass will necessarily have different metabolic demands from quiescent cells, who may require lower levels of nucleotide and protein synthesis. Likewise, cells of different lineages will have cell-type-specific functions, which may increase reliance on certain metabolic pathways, or reside in unique microenvironments that dictate nutrient availability. Beyond meeting cell demands, metabolites also serve regulatory roles, influencing signaling pathways and chromatin modifications that ultimately modulate gene expression programs. Collectively, cell state, lineage, and location collaborate to determine the metabolic preferences and requirements of a given cell. Underscoring the importance of metabolic fine tuning to meet cell-type-specific functions, metabolic dysregulation is causative in developmental pathologies such as inborn errors of metabolism, malignancy, and other diseases.[2]

In this review, we will examine the hypothesis that each major function of stem cells—self-renewal, differentiation, quiescence, and survival in the stem cell niche—is closely tied to cell metabolism. As a result, different stem cells can exhibit vastly different metabolic features, and metabolic perturbations can affect outcomes of specific stem cell populations. Accordingly, there are no universal hallmarks of stem cell metabolism; rather, metabolic signatures of stem cells are as varied as the identities and functional states of stem cells themselves. For example, embryonic stem cells isolated from the inner cell mass of the pre-implantation blastocyst are highly proliferative and undergo rapid cell division both in the pluripotent state and upon the induction of multilineage differentiation. Tissue-resident stem cells found in the intestine are maintained in both highly proliferative and more dormant, metabolically quiet states. In other tissue compartments such as the skin, muscle, brain, and hematopoietic system, typically quiescent stem cell populations are rapidly mobilized in response to normal physiologic cues or in the context of tissue regeneration. The balance between proliferation and quiescence will invariably affect the metabolic signatures of different stem cell populations.

Stem cells must make this choice between proliferation and quiescence while retaining the capacity to form all cell types of their respective lineage. Thus, the metabolic programs that support quiescence or proliferation must be compatible with the metabolic programs that tune gene expression programs to control stem cell fate. Consistent with their role as reservoirs to repopulate tissue compartments, stem cells also have metabolic strategies to respond to cellular stressors and resist cell death. Finally, stem cells must operate within and respond to metabolic cues from their endogenous niche. All together, these considerations ensure that there is no unified formula for stem cell metabolism; rather, cell metabolism is harnessed to support specific needs of different stem cell populations. Below, we discuss approaches for studying stem cell metabolism and integrate recent work to generate an overview of how metabolism supports five hallmarks of stem cell function.

TOOLS FOR STUDYING STEM CELL METABOLISM

As technological advances have improved our ability to isolate and study stem cell populations, numerous studies have begun

to examine metabolism in these cells. However, it is important to acknowledge several key challenges in studying stem cell metabolism.[3] Stem cells are rare cell states that are proportionally limited relative to their differentiated progeny. Most metabolomic studies require large numbers of cells, which constrains their application to these cell populations. While stem cells can be enriched from tissues by fluorescence-activated cell sorting, these approaches can significantly alter the cellular metabolome.[4,5] An alternative approach is to isolate cells and then allow them to recover, followed by profiling *ex vivo*.[6] Here, results may be confounded by the changing conditions, as tissue-resident stem cell identity and metabolism are coupled to their physiological niche,[7] which is difficult to recapitulate *in vitro*. Nevertheless, both approaches represent valuable starting points for generating testable hypotheses regarding metabolic pathways that may be active or important in specific cell types. Genetic or pharmacologic manipulation of specific metabolic nodes is required to distinguish between metabolic signatures that correlate with specific cell states (for example, nutrient or pathway preferences) from those that are truly required for establishing or maintaining specific cell identities.

While metabolomic studies are limited by cell number, new sequencing approaches allow the interrogation of gene expression at the single-cell level. Extensive transcriptomic characterization of stem cell populations has demonstrated that metabolic genes are differentially expressed in stem cells relative to their activated counterparts.[8–11] It is critical to note, however, that levels of metabolic gene transcripts are not sufficient to support claims regarding metabolic pathway engagement.[12,13] Transcript levels imperfectly predict metabolic enzyme levels,[14] as protein synthesis, post-translational modifications, or turnover can each be regulated by separate inputs. Further, metabolic flux is often driven by changes in metabolite supply or demand, and flux is not necessarily constrained by levels of specific enzymes.[15] Finally, in addition to being the driver of increased metabolic flux, RNA can also signal a compensatory response to decreased metabolic pathway activity, as for example when AMPK activates mitochondrial biogenesis in response to compromised energy production.[16]

How then can we measure metabolism directly in stem cells? One popular approach has been extracellular flux assays (EFA), which measure either the oxygen consumption rate (OCR) or extracellular acidification rate (ECAR) of isolated cells and require comparatively less starting material than other metabolomic measurements. These values can correlate with oxidative phosphorylation and glycolytic activity, respectively, but these values are not absolute quantifications of either pathway activity.[17] A high ECAR does not mean a given cell type is purely glycolytic and does not engage in oxidative phosphorylation. Neither pathway occurs in mutual exclusivity from the other; oxidative phosphorylation is dependent on the upper steps of glycolysis that generate pyruvate from glucose to fuel the TCA cycle.[18] While EFA can be used to probe differential metabolic pathway engagement between samples, care should be taken not to ascribe absolute metabolic activity based on these results.

More information can be derived by directly measuring metabolites themselves,[19,20] though steady-state metabolite levels must be taken in context, as increased levels of pathway intermediates can indicate increased flux or a block later in the pathway. Direct assessment of metabolic flux requires the tracing of labeled substrates into downstream metabolic networks coupled with analysis at subsequent timepoints and/or information about nutrient consumption rates to build quantitative models of metabolite usage in different biochemical pathways.[21] In the absence of a full flux analysis, tracing with heavy labeled substrates allows for determination of which cellular metabolites are derived from a given (labeled) substrate in various cell types. Labeling patterns may also provide additional information about the biochemical route from substrate to metabolite of interest.[22] The application of these approaches has been limited by technical challenges in most rare stem cell populations *in vivo*. One promising approach is to assess isotope incorporation into metabolic endpoints (DNA, protein), as these readouts are more stable over time and less susceptible to the issues introduced during cell sorting.[23] For these reasons, stem cells that can be readily expanded *in vitro* in the presence of specialized media formulations optimized for isotope tracing provide an experimentally tractable system for assessing metabolic networks during cell fate transitions.

Metabolic measurements are best when complemented by studies examining the functional consequences of metabolism to stem cell identity. Among the most powerful approaches for studying stem cell metabolism is the direct perturbation of metabolic genes in specific stem cell compartments. While all the approaches discussed above have utility for studying stem cell metabolism, loss-of-function studies such as these are powerful because of their ability to illustrate stem-cell-specific metabolic requirements. To the extent that different stem cell types require similar metabolic pathways, these approaches allow us to define common themes in stem cell metabolism and facilitate our understanding of how metabolism supports different cell states in both normal development and tissue regeneration.

Despite the challenges of studying stem cell metabolism, the field has made enormous strides uncovering general paradigms for how metabolic networks support specific stem cell functions. We describe below aspects of stem cell metabolism that drive or enable five core hallmarks of stem cell biology (Figure 1).

HALLMARK 1: METABOLISM SUPPORTS STEM CELL PROLIFERATION

The metabolic pathways that support mammalian cell proliferation have been extensively studied. Much of this work has been done in cancer cell lines, and the role of metabolism in cancer cell proliferation has been reviewed elsewhere.[24] Like cancer cells, a key feature of stem cells is the capacity for rapid proliferation. Both cell types have therefore evolved metabolic strategies to support proliferation, many of which are shared between stem cells and cancer cells.[25] Continuous cell proliferation requires a net increase in biomass; accordingly, nutrients must be taken up and catabolized to generate the building blocks for macromolecules such as proteins, nucleic acids, and cellular membranes. Much work has focused on the use of glucose and glutamine to fuel proliferation, as these are the two most abundant nutrients in mammalian cell culture. However, the specific metabolic pathways used to acquire nutrients and meet cellular anabolic demands can vary between cell types and are dynamically altered in cellular differentiation.

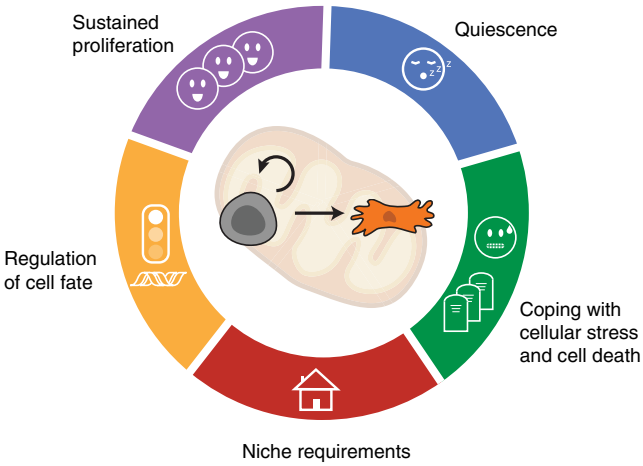

Figure 1. Metabolic support of stem cell functions
While metabolism can vary dramatically between different stem cell types, several common themes have emerged in stem cell metabolism that support vital cellular functions. Specific metabolic strategies support stem cell proliferation (1) and quiescence (2). Metabolism is likewise involved in stem cell responses to cellular stressors and cell death (3). By influencing the deposition or removal of epigenetic modifications, metabolism can also control stem cell fate (4). Stem cells are also influenced by the metabolic requirements of their endogenous niche (5).

Aerobic glycolysis and respiration

Glucose is an essential nutrient for most cells, as a source of energy and for its role providing the sugar backbone and reducing equivalents required for macromolecule synthesis (Figure 2A). Glucose is metabolized via glycolysis to form pyruvate, and glycolytic intermediates serve key roles in proliferating cells. Glucose diversion to the pentose phosphate pathway yields ribose for nucleotide synthesis and/or NADPH. Pyruvate can enter the tricarboxylic acid (TCA) cycle, where complete combustion yields reducing equivalents (NADH, $FADH_2$) that fuel oxidative phosphorylation through the electron transport chain (ETC). Nevertheless, most rapidly proliferating cells paradoxically throw away most of their glucose-derived carbons as lactate, even when supplied sufficient oxygen to sustain TCA cycle flux.[26] This process of aerobic glycolysis, often referred to as the "Warburg effect," can net generate only 2 molecules of ATP per molecule of glucose. Aerobic glycolysis was discovered in cancer cells and was long considered to be a metabolic quirk of transformed cells. Studies in stem cells challenged this model, revealing that many stem cell populations also exhibit aerobic glycolysis, which appears to be a feature of most rapidly proliferating cultured cells.[18] Why rapidly proliferating cells discard carbon as lactate is largely an open question, although recent evidence indicates that the conversion of pyruvate to lactate via lactate dehydrogenase (LDH) is favored when the demand for NAD^+ regeneration outstrips the demand for ATP.[27,28]

Given the importance of glucose metabolism, many studies have investigated the role of glycolysis and the fate of pyruvate in a variety of cell types. In stem cells, the fate of pyruvate—to be excreted as lactate or burned in the TCA cycle—is closely tied to cell state. One example of this is PSCs, which are transcriptionally reminiscent of the pre- or post-implantation epiblast, depending on their *in vitro* culture conditions.[29,30] The addition of inhibitors against MEK and GSK3β is sufficient to drive mouse PSCs into the naive state of pluripotency reminiscent of the pre-implantation epiblast[31]; further interventions are required to capture the naive pluripotent state in human PSCs.[32,33] In general, PSCs cultured in the naive state have higher basal respiration relative to their post-implantation counterparts,[34–37] and naive mouse PSCs preferentially incorporate glucose-derived pyruvate into TCA cycle intermediates.[38–40] Increased respiration enables optimal proliferation in naive ESCs and may even be required for entry into the naive pluripotent state,[36] but the precise outputs of respiration that benefit naive PSCs remain unknown. Notably, high respiration is not mutually exclusive with aerobic glycolysis: some naive human PSCs also exhibit high lactate production, reflecting overall elevated glucose metabolism and/or oxidation of additional substrates in these cells.[34]

As PSCs exit the naive pluripotent state, they transiently decrease glucose oxidation in the TCA cycle and increase lactate secretion.[34,36,40,41] Accordingly, PSCs mimicking post-implantation epiblast states are glycolytic with minimal glucose oxidation. This aerobic glycolysis phenotype is reversed upon further differentiation, which is often accompanied by decreased glucose flux through glycolysis and decreased lactate secretion.[34,41,42] Aerobic glycolysis is not universally downregulated during differentiation, however; human PSCs maintain high glycolytic flux when differentiated to ectoderm, but not mesoderm and endoderm.[43] As with studies of cancer cells, whether aerobic glycolysis confers specific benefits to stem cells remains an open question. Certainly, pharmacologic inhibition of glycolysis impairs proliferation in several PSC models.[34,44] Likewise, how such major metabolic shifts are coordinated is not fully understood, although several pluripotency transcription factors are directly implicated in metabolic control. For example, OCT4 and MYC directly regulate glycolytic genes,[43,45] and STAT3 is reported to translocate to the mitochondria to activate mitochondrial gene expression.[36]

The relative downregulation of glycolysis with PSC differentiation stands in contrast to some tissue-resident stem cells where glycolysis is associated with stem cell activation and differentiation to progenitors. In hematopoietic stem cells (HSCs), quiescent HSCs exhibit lower expression of glycolytic genes, have a lower ECAR, and are less sensitive to pharmacologic inhibition of glycolysis *in vitro*.[46] As HSCs activate and differentiate, reliance on glycolysis increases. Genetic deletion of LDHA or the upstream glycolysis enzyme PKM2 decreased lactate production and compromised hematopoiesis during forced hematopoietic system reconstitution in transplantation assays, without affecting maintenance of homeostatic HSCs.[47] Similarly, deletion of PDK2 and PDK4 or silencing of PDK1, which would be expected to promote mitochondrial pyruvate oxidation at the expense of glycolytic flux, impairs HSC reconstitution capacity.[48,49] Importantly, it is unclear whether these defects observed in glycolysis-deficient HSCs are due to decreased glycolysis or if enhanced pyruvate oxidation and concomitant increased respiration reduces self-renewal potential. Increased mitochondrial respiration can generate reactive oxygen species (ROS) that impair HSC function.[50] Consistently, antioxidant treatment decreased ROS levels and partially rescued bone marrow transplantation in LDHA-deficient HSCs, suggesting that decreased oxidative

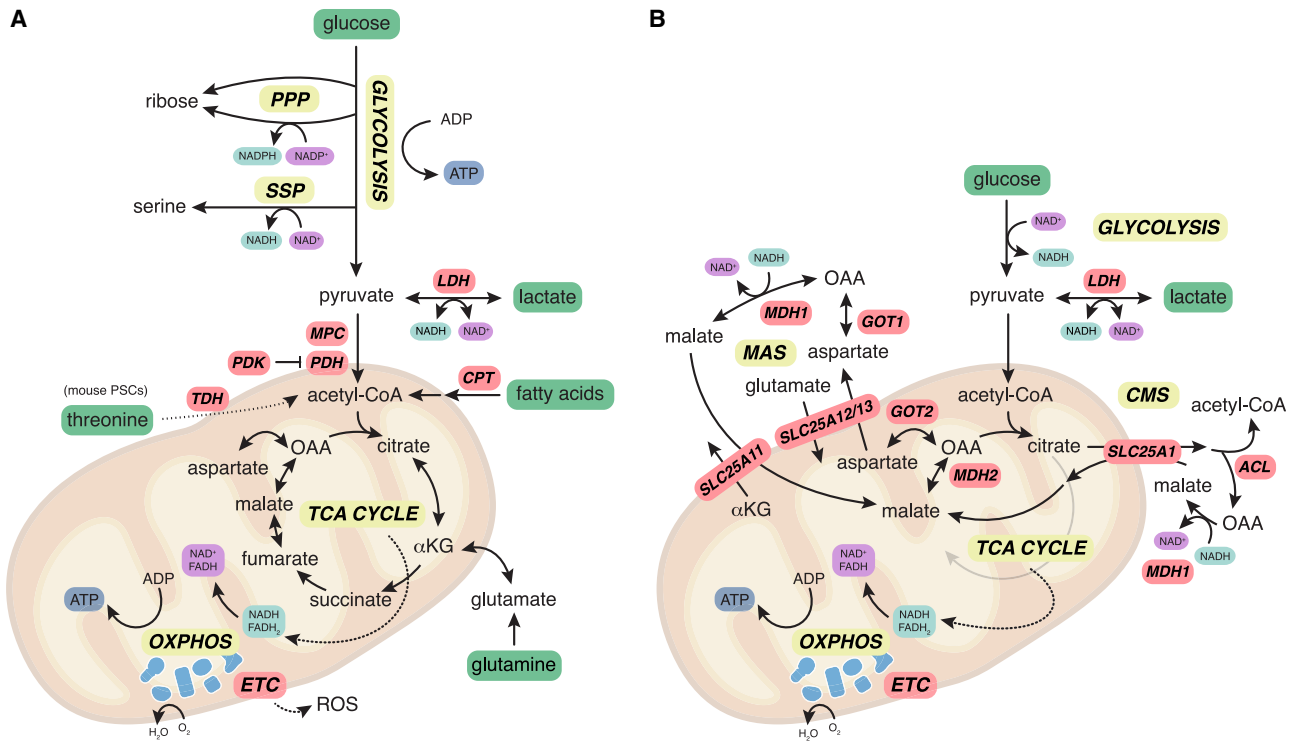

Figure 2. Metabolic pathways supporting stem cell proliferation

Multiple metabolic pathways generate the building blocks required for stem cell proliferation.

(A) Diverse substrates fuel anabolic reactions that generate either metabolic intermediates or reducing equivalents (NADPH, NADH, $FADH_2$) for macromolecule synthesis. One major substrate is glucose, which is converted to pyruvate by glycolysis. Pyruvate either can be converted to lactate by LDH and discarded from the cell or can enter the TCA cycle where it is oxidized and generates reducing equivalents that power oxidative phosphorylation by the ETC. Regulation of oxidative metabolism is key to both PSC and tissue stem cell function. PPP, pentose phosphate pathway; SSP, serine synthesis pathway.

(B) Many cellular oxidation reactions—including the oxidation of glucose—require NAD^+ as a cofactor. Therefore, multiple metabolic pathways have evolved to regenerate NAD^+ from NADH. Differential modes of NAD^+ regeneration by either the LDH reaction or by cross-compartmental electron shuttles underlie stem cell identity. MAS, malate-aspartate shuttle; CMS, citrate-malate shuttle.

phosphorylation might play a protective role in the stem cell compartment by limiting ROS production.[47]

Accordingly, activated HSCs have both a higher OCR than quiescent HSCs and exhibit increased mitochondrial membrane potential.[46,51,52] Glucose consumption and tracing to TCA cycle intermediates also increases with HSC commitment to differentiated progenitors *in vivo*, supporting a model where HSC differentiation is accompanied by both increased glycolytic and oxidative pyruvate metabolism.[46,53] Despite this increase in pyruvate oxidation with HSC commitment, blockage of pyruvate entry into the TCA cycle by deletion of PDHA1 had no effect on bone marrow hematopoietic development.[53] Nevertheless, substrate oxidation via the TCA cycle likely remains critical, as loss of the mitochondrial respiratory chain complex III subunit Rieske iron-sulfur protein led to HSC cell cycle exit and death.[54] Therefore, some level of mitochondrial respiration is required for HSC maintenance.

Manipulating mitochondrial pyruvate entry alters stem cell activity in other tissue compartments. The mitochondrial pyruvate carrier MPC1 is dynamically regulated in neural stem cells (NSCs), and *in vitro* and *in vivo* genetic ablation of MPC1 leads to NSC activation.[55] Similarly, deletion of MPC1 in intestinal stem cells (ISCs) leads to increased stem cell proliferation *in vivo* and *in vitro* organoid formation.[56] In hair follicle stem cells (HFSCs), deletion of MPC1 accelerates the hair cycle and is accompanied by increased lactate formation consistent with higher pyruvate flux through LDHA.[57] However, the precise mechanisms by which MPC1 deletion promotes stem cell activation in these cell types—for example, whether such phenotypes arise from impaired mitochondrial pyruvate oxidation or as a result of increased usage of pyruvate for alternative fates—remains unknown.

Regulation of cofactor redox

Why does it matter whether cells oxidize pyruvate in the TCA cycle or convert pyruvate to lactate through aerobic glycolysis? A major distinction centers on the fate of NAD^+, a critical cofactor that is reduced to NADH during cellular oxidation reactions that sustain cell proliferation[58] (Figure 2B). Each molecule of pyruvate generated from glucose requires the reduction of NAD^+ to NADH. In most rapidly proliferating cells, NADH is converted back to NAD^+ via lactate dehydrogenase; accordingly, NAD^+ demand can drive aerobic glycolysis.[27] Consistent with the role of aerobic glycolysis in sustaining stem cell proliferation, loss of LDHA results in delayed HFSC activation during the normal hair cycle[57] and impaired long-term bone marrow transplantation by HSCs but not normal hematopoietic development.[47] In epidermal stem cells *in vitro*, mitochondrial pyruvate oxidation

induces dependence on exogenous serine, as *de novo* serine synthesis depends on cytosolic NAD^+.[59] These studies suggest that NAD^+ regeneration may be limiting for stem cells in settings of increased metabolic demand.

However, the conversion of pyruvate to lactate is net redox neutral as NAD^+ is required to generate pyruvate from glucose during glycolysis. Therefore, additional routes of NAD^+ regeneration are necessary. In the mitochondria, a major route for NAD^+ regeneration from NADH is complex I of the electron transport chain (ETC). Continuous NAD^+ regeneration via complex I of the ETC sustains pyruvate oxidation in the mitochondria, which generates several molecules of mitochondrial NADH through the reactions of the TCA cycle.[26] Cytosolic NADH cannot be directly transported into mitochondria; rather, the reducing equivalents are transferred to metabolites that, when imported into mitochondria, reduce NAD^+ to NADH to complete the effective electron transport. Such "electron shuttles" coordinate cross-compartment metabolic communication and include the glycerol 3-phosphate shuttle, the malate-aspartate shuttle, and the citrate-malate shuttle.[60]

Despite the importance of cytosolic NAD^+ regeneration, relatively little is known about how electron shuttles are regulated in different cell types. Recent studies demonstrated that the citrate-malate shuttle is engaged in both cancer cell lines and PSCs in culture.[40] In PSCs, citrate-malate-shuttle utilization increases during the exit from naive pluripotency, and loss of citrate-malate shuttle activity is incompatible with differentiation.[40] Components of the citrate-malate shuttle are also required for hematopoietic regeneration following myeloablation[61] and promote the osteogenic capacity of mesenchymal stem cells *ex vivo*.[62] These studies raise the possibility that the method of NAD^+ regeneration might play a critical role in supporting stem cell proliferation or identity.

Substrate preferences

While considerable attention has been given to glucose, other substrates also contribute as major energy sources to fuel metabolic pathways for cellular anabolism. For example, glutamine oxidation is a major source of TCA cycle intermediates, supporting both fueling respiration and generating the metabolic precursors required to support cell growth *in vitro*.[60] Not surprisingly, glutamine supplementation is required for the *in vitro* culture of almost all mammalian cells.[63,64] A notable exception to this rule is naive PSCs, which can proliferate in the complete absence of exogenous glutamine.[33,38,39] This glutamine independence is enabled by reduced glutamine catabolism: cells cultured in the ground state of naive pluripotency incorporate lower levels of glutamine into TCA cycle intermediates, allowing cells to maintain production of key biosynthetic intermediates in the absence of exogenous glutamine.[38] Because naive PSCs with the highest self-renewal reduce dependence on exogenous glutamine, glutamine withdrawal selects for naive PSCs with the highest propensity for self-renewal within a heterogeneous population of PSCs.[39] Like more committed human PSCs, the developing mesoderm and endoderm continue to rely on glutamine as a metabolic substrate, but the developing ectoderm does not; rather, glutamine serves as a signal to activate mTORC1-dependent differentiation in this lineage.[65] These results illustrate the large changes in nutrient preferences that accompany PSC development and further underscore the cell-type-specific role of nutrients as substrates for growth and inputs to signaling pathways that regulate cell proliferation.

In contrast to glutamine, which is dispensable for the proliferation of naive mouse and human PSCs, threonine is required by naive mouse PSCs.[66,67] Mouse PSCs express the enzyme threonine hydrogenase (TDH) which catabolizes threonine to glycine and acetyl-CoA to fuel one-carbon metabolism and the TCA cycle. Notably, TDH is uniquely expressed in mouse PSCs and is considered non-functional in human cells.[68] Methionine, an essential amino acid that also supplies one-carbon metabolism, is likewise required for growth of PSCs and most cultured cells.[69] As most amino acids contribute to cellular processes beyond protein synthesis, it is likely that changes in use of many amino acids may accompany stem cell activation and differentiation. How cells acquire and consume amino acids, and whether this is tied to proliferation of specific stem cell compartments, is largely unexplored.

Other fuel sources such as lipids can support cell growth by directly contributing to lipid pools required for cell membranes or by serving as a fuel source to support cellular bioenergetics. Exogenous lipids are readily acquired by PSCs,[70] and lipid availability can influence PSC fate, in part by controlling signaling pathways.[71] *De novo* lipid synthesis is especially important for culture of cells in serum-free formulation, as serum provides a robust source of exogenous lipids.[70] Even *in vivo*, *de novo* lipid synthesis is critically important in certain stem cell populations, highlighting the importance of biosynthetic pathways to support stem cell expansion.[72] Fatty acids are also consumed as fuel for metabolic networks, notably in HSCs and ISCs where fatty acid oxidation is required for stem cell maintenance and self-renewal *in vivo*.[73,74] Consistently, deletion or pharmacologic inhibition of CPT1a—the rate-limiting enzyme in fatty acid oxidation—impairs stem cell maintenance in these compartments.[73,74] Whether fatty acid oxidation plays a key bioenergetic role or provides other signals to promote stem cell function remains to be determined. All together, these studies demonstrate that stem cells in different states of self-renewal or activation can dramatically rewire their substrate preferences, and future work will aim to unravel what specifies the substrate preferences in each cell state and how these metabolic preferences contribute to functional outcomes of different stem cell populations.

HALLMARK 2: METABOLIC ADAPTATIONS DURING QUIESCENCE

In addition to supporting the demands of cellular proliferation, metabolism can support the maintenance of non-dividing stem cell states. Many tissue stem cells undergo periods of quiescence, which is defined as temporary cell cycle exit into the G_0 state that can be reversed in response to external stimuli.[75] Tight control over quiescence is critical for normal stem cell homeostasis, and inappropriate proliferation can lead to exhaustion of the stem cell compartment. While the definition, transcriptional regulation, and cellular function of quiescence can vary considerably among different tissue stem cell types,[76,77] several common metabolic features of quiescent cells have emerged.

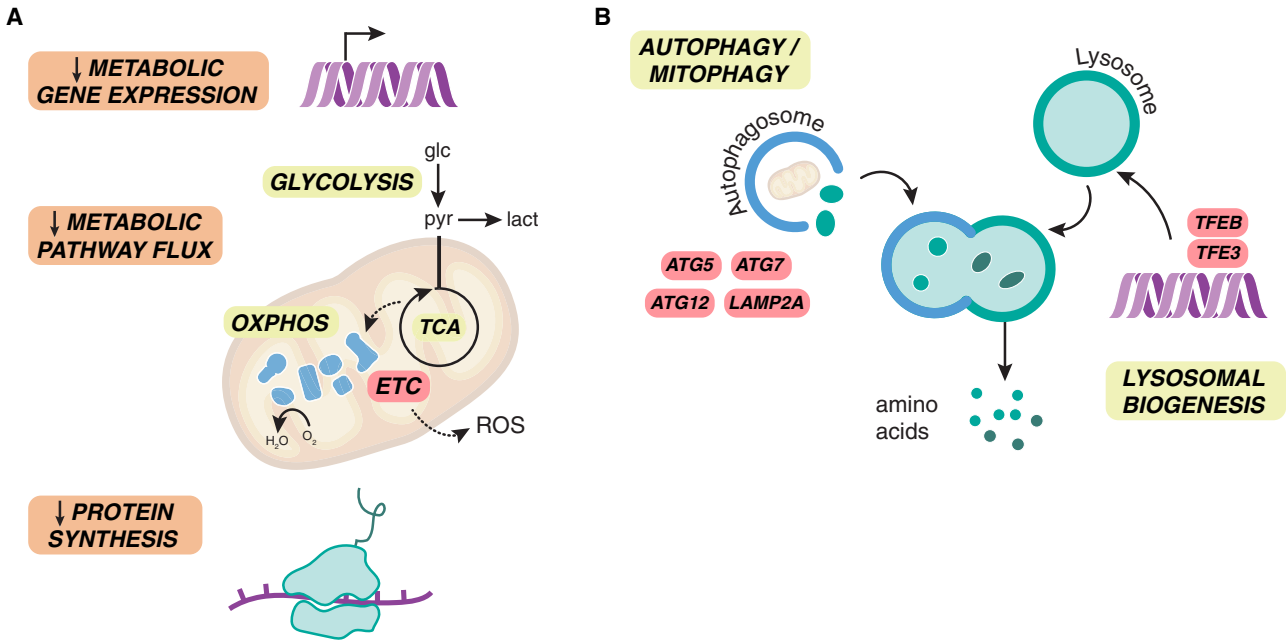

Figure 3. Metabolic strategies to support stem cell quiescence

Cell cycle exit in quiescence is supported by specific metabolic rewiring. Decreased metabolic demand (A) is achieved by altered metabolic gene expression, decreased metabolic pathway flux, and an overall reduction in metabolic outputs such as protein synthesis. In concert with decreased metabolic demand, increased cellular recycling processes (B) such as autophagy and lysosomal biogenesis liberate metabolites to support cellular bioenergetics and constrain mitochondrial respiration via mitophagy.

Lower metabolic activity

Consistent with cell cycle exit, quiescence has historically been considered a metabolically inert state, where decreased cellular proliferation is thought to be accompanied by decreased demand for the products of anabolic metabolism (Figure 3A). Early evidence for the concept of metabolic quiescence was derived from sequencing studies that profiled quiescent versus activated stem cells across multiple tissue types.[78] Quiescent stem cells consistently show altered expression of genes involved in multiple metabolic processes relative to their proliferative counterparts.[8–11] In NSCs specifically, quiescence is marked by high expression of glycolytic and fatty acid oxidation genes that are downregulated upon activation with a concomitant increase in mitochondrial and oxidative phosphorylation gene signatures.[10,11]

While numerous studies have noted shifts in metabolic gene expression during the acquisition of and exit from quiescence, relative few studies have functionally characterized metabolic changes between these states. Technical challenges in isolating sufficient volumes of quiescent stem cells for metabolomic studies have limited their interrogation. Experiments performed in HSCs have confirmed that quiescence is accompanied by low mitochondrial activity and low reliance on glycolysis upon *ex vivo* culture in this lineage.[46] Genetic interventions that increase mitochondrial substrate oxidation result in cell cycle activation *in vivo* and loss of HSC quiescence.[48] Paradoxically, deleting an essential subunit of ETC complex III also leads to loss of HSC quiescence *in vivo*.[54] In muscle stem cells (MuSCs), stem cell activation is accompanied by mitochondrial fragmentation and elevated mitochondrial ROS production that triggers cell cycle entry. Accordingly, excessive fragmentation via genetic loss of the mitochondrial fusion protein OPA1 causes premature MuSC activation and depletion from the stem cell pool.[79] These studies suggest that altered metabolic activity not only accompanies quiescence but might play a regulatory role in its maintenance, although the specific metabolic trigger(s) that control quiescence remain to be determined.

Which metabolic outputs are truly reduced during quiescence is largely an open question, but many studies point to reduced protein synthesis as a common feature of quiescent stem cells.[80] Decreased translation—among the most costly bioenergetic cellular processes—has been observed in PSCs,[81,82] HSCs,[83] NSCs,[84] HFSCs,[85] and MuSCs.[86] Across these cell types, the specific benefit of low translation for quiescence remains unclear.[80] However, forced protein synthesis by genetic perturbations causes stem cells to exit quiescence and activate at the expense of self-renewal, leading to the eventual depletion of the stem cell pool.[83,85,86] Conversely, decreased protein synthesis is compatible with HSC maintenance but prevents HSC activation and regenerative capacity.[83] These studies illustrate how synchronicity of cellular functions with metabolic state is critical for stem cell maintenance, and it will be interesting for future studies to investigate whether changes in protein synthesis are sufficient to explain metabolic quiescence or if depression of other metabolic programs is likewise required for maintenance of quiescent stem cells.

In contrast to adult stem cells that alternate between periods of quiescence and activation, embryonic stem cells capture a developmentally transient state in which epiblast cells undergo a single period of rapid development to form the developing blastocyst—a process that can be reversibly halted during embryonic diapause.[87] Diapaused embryos are thought to exhibit

lower usage of metabolic pathways, consistent with a more quiescent state.[87] However, specific metabolic changes in diapause are poorly characterized, and opposing studies have reported both increased and decreased glycolysis[88,89] and fatty acid oxidation.[88,90] Diapaused embryos are reported to increase reliance on lipid droplets[91] and autophagy[92] for survival, suggesting a potential change in substrate preferences during this cell state. Importantly, many of the cues that regulate diapause in embryos are metabolic. Reductions in polyamine levels can delay blastocyst implantation[93] and glucose withdrawal delays the reentry from diapause,[94] but how these nutrients fluctuations are communicated to the embryo to dictate cell fate remains unclear.

Given the technical limitations of studying embryonic metabolism *in vivo*, considerable effort has gone into developing *in vitro* models of diapause-like states. Two distinct approaches—inhibition of the transcription factor MYC and inhibition of mechanistic target or rapamycin (mTOR)—are sufficient to induce the reversible pausing of cultured PSCs.[95,96] Transcriptionally, these paused PSCs resemble diapaused epiblasts and share common features of diapause including decreased transcription, RNA splicing, and protein synthesis. While modulation of these pathways is sufficient to induce diapause-like states,[97] whether endogenous regulation of these pathways regulates diapause in developing embryos remains unclear.[96] Altogether, these orthogonal *ex vivo* pausing strategies have yielded considerable insight into the biology of diapause, and the development of other systems will likely increase our understanding of this unique metabolic state.

Increased autophagy

Regulation of macromolecule synthesis is balanced with cellular catabolism (Figure 3B). In addition to decreased expression of metabolic genes, quiescent stem cells have higher expression of genes involved in autophagy.[78] Autophagy is a process of intracellular "self-eating" that targets cytoplasmic cargo to the lysosome for degradation. In quiescence, autophagy—while incompatible with net biomass generation—might liberate nutrients that allow cell survival while in a metabolically low or isolated state. Similarly, autophagy can be utilized by non-dividing cells to remove dysfunctional organelles or cellular debris that might otherwise be passed along to dividing progeny.

The best examination of autophagy in quiescence are studies where components of the autophagosome are genetically perturbed in the stem cell compartment. Conditional loss of the essential autophagy gene ATG7 in the fetal hematopoietic system led to decreased numbers of HSCs that were unable to reconstitute the hematopoietic compartment.[98] Loss of autophagy in adult HSCs by deletion of ATG12 likewise impaired HSC self-renewal and regenerative capacity.[99] In both settings, defective autophagy led to mitochondrial accumulation and increased oxygen consumption, suggesting that one role of autophagy in HSCs is mitochondrial clearance through mitophagy to constrain metabolic capacity. Similar phenotypes were observed upon loss of ATG5[100] or ATG7[101] in ISCs. Autophagy-deficient ISCs have increased levels of ROS, undergo higher levels of apoptosis, and have impaired regeneration capacity following irradiation. These phenotypes were partially reversed upon antioxidant treatment, implying that autophagy constrains oxidative stress in ISC homeostasis.

An emerging question is how autophagic cargo are selectively targeted for degradation to maintain stem cell quiescence. In chaperone-mediated autophagy, phosphorylated or acetylated amino acid motifs mark specific proteins for lysosomal degradation. Abrogation of chaperone-mediated autophagy by deletion of LAMP2A in HSCs *in vivo* results in the buildup of dysfunctional proteins involved in glycolysis and fatty acid oxidation.[102] Consequently, LAMP2A-deficient cells are unable to increase glycolytic activity and lipid metabolism upon HSC activation and are unable to repopulate the bone marrow following myeloablation and in serial transplantation.[102] In PSCs, chaperone-mediated autophagy increases with PSC differentiation and targets mitochondrial proteins to suppress pluripotency.[103]

Autophagy is also required in MuSC activation *ex vivo*, where it has been proposed to provide the nutrients necessary for increased anabolic demands.[104] Additional *in vivo* studies have demonstrated that ATG7 is required for maintenance of the quiescent MuSC pool by preventing MuSC senescence,[105] suggesting that autophagy likely plays multiple roles even within the same stem cell population. Interestingly, autophagy has been shown to decline with age, and decreased levels of autophagy might contribute to the reduction in tissue stem cell activity seen with aging. Across multiple tissues, autophagy-deficient stem cells share features with their older normal counterparts,[99,102,105] reflecting shared biology between autophagy loss and aging and suggesting that augmentation of autophagic pathways might be beneficial for the function of these cells with aging.

Lysosomal biogenesis

Lysosomes—the end destination of autophagic cargo—are an additional point of metabolic regulation in quiescent stem cells. Quiescent HSCs have higher numbers of enlarged lysosomes[46] and increased nuclear localization of the lysosomal biogenesis transcription factor TFEB.[106] TFEB overexpression in HSCs *ex vivo* is sufficient to impose a pro-lysosomal transcriptional program that restricts HSC cell division.[106] Conversely, TFEB knockdown increased numbers of cycling HSCs and decreased HSC engraftment in transplantation assays.[106] Lysosomal activity in HSCs is additionally regulated by asymmetric cell division. Increased lysosomal inheritance by daughter cells correlates with lower mitochondrial membrane potential and quiescence, and daughter cells that inherit fewer lysosomes are more likely to differentiate to progenitors.[107] While the mechanism by which lysosomes promote HSC quiescence remains unclear, these studies suggest that one function of lysosomes in HSCs might be to restrict levels of metabolic proteins or organelles that enable differentiation; alternatively, lysosomes serve as signaling hubs[108] and in this regard may directly control stem cell fate.

Quiescent NSCs are similarly characterized by enlarged lysosomes that actively degrade protein aggregates.[109,110] The induction of NSC quiescence is accompanied by increased lysosomal proteolysis; conversely, inhibiting lysosomal acidification leads to the accumulation of protein aggregates and NSC proliferation.[109,110] Expression of the lysosomal biogenesis factor TFEB *ex vivo* led to NSC quiescence and decreased levels of protein aggregates.[109,110]

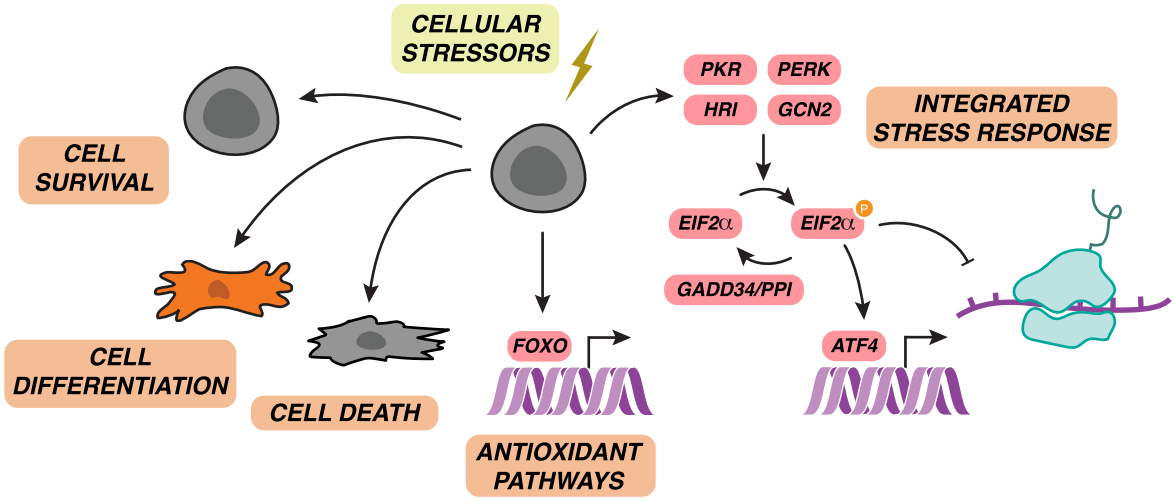

Figure 4. Stem cell responses to cellular stress

Stem cells have distinct responses to cellular stress. While PSCs are resistant to cell death, tissue stem cell populations have increased propensity for cell death or differentiation in response to external stressors. Stem cells also have increased reliance on cellular detoxification and antioxidant pathways. Activation of the integrated stress response has also been linked to stem cell function and dysfunction in normal development and disease.

High lysosomal activity also defines naive PSCs; here, lysosomal activity licenses hydrolysis of extracellular protein and enables amino acid independence.[111] Suggestively, the TFEB homolog TFE3 is sufficient to trap PSCs in the naive, ground state of pluripotency,[112,113] although the degree to which the effect of TFE3 on PSC fate requires lysosomal hydrolysis or induction of quiescence-like features remains to be determined. While the precise benefit of increased autophagic and lysosomal flux is likely context specific, these studies demonstrate the importance of cellular degradation pathways in maintaining the quiescent state and during stem cell activation.

HALLMARK 3: RESISTANCE TO METABOLIC STRESS AND CELL DEATH

Metabolic support of cell survival and proliferation is balanced with tight metabolic control of cell death.[114] As reservoirs for tissue regeneration in settings of injury, stem cells must be able to withstand some degree of cellular stress. Strikingly, viable tissue stem cells capable of engraftment have been isolated up to four days postmortem,[115] suggesting that stem cells are uniquely positioned to resist external stressors. At the same time, precautions are taken to ensure defective stem cells do not contribute to the progenitor pool and so appropriate control of cell death programs is critical for normal development and tissue homeostasis.[116] The diverse strategies that govern stem cell responses to stress are an area of active inquiry (Figure 4).

Regulation of cell death

Rapid proliferation in PSCs is balanced with tight control of cell death pathways. In the developing mouse embryo, the epiblast cell number increases 100-fold from E5.5 to E7.5.[117] Concomitantly, large numbers of cells are eliminated through apoptosis at E6.5.[118] This induction of cell death with differentiation has been further studied in connection with sensitivity to ionizing radiation. Low-dose (0.5 Gy) irradiation induces apoptosis of E6.5 embryos, but not their pre-implantation counterparts that have not yet begun gastrulation.[119] This differential resistance to cell death has also been observed in PSCs in culture and extended to other cellular stresses. For example, loss of microRNAs, oxidative stress, and endoplasmic reticulum (ER) stress all lead to apoptosis induction in EpiSCs but not in their naive counterparts.[120,121]

These results have led to the hypothesis that embryonic differentiation is accompanied by state-specific propensity for cell death. Human PSCs which *in vitro* are maintained in a primed, post-implantation-like state are more sensitive to DNA damage and ER stress than differentiated cells.[122,123] Consistently, the PSC stage with the highest priming for cell death corresponds to the embryonic stage where apoptosis occurs,[118] illustrating the utility of cell culture models to interrogate key regulatory steps during normal development. As apoptosis is tightly linked to mitochondrial permeabilization and caspase activation,[124] several studies have examined the role of mitochondrial dynamics in PSCs. Naive PSCs have higher expression mitochondrial fusion regulator DRP1, and loss of DRP1 is sufficient to lower the apoptotic threshold in these cells. Conversely, overexpression of DRP1 can inhibit apoptosis at baseline and in response to endoplasmic reticulum and oxidative stress in differentiating PSCs.[121] Mitochondrial dynamics therefore might not only support bioenergetics but also play an important role in promoting PSC resistance to stress.

In contrast to PSCs that represent transient developmental states, tissue stem cells function over organismal lifetimes and must guard against accumulation of damaged organelles or compromised genetic integrity. Accordingly, tissue stem cells often exhibit increased propensity for cell death relative to their more committed counterparts. For example, following irradiation, HSCs exhibit higher levels of apoptosis than hematopoietic progenitors.[125] Enforced oxidative DNA damage or ER stress similarly reduced HSC proliferation and viability

relative to more differentiated cells.[126,127] In response to oxidative stress, MuSCs fuse with proximal myofibers, thereby removing damaged MuSCs from the stem cell compartment.[128] Stem cells also cope with damage and exit the stem cell pool through differentiation. In the epidermis, differentiation is a common response to stresses such as UV radiation or oncogene expression.[129–131] Similarly, ER stress in ISCs both *in vitro* and *in vivo* led to stem cell differentiation with decreased expression of normal ISC markers.[132] These responses allow for maintained barrier function while preserving the integrity of the stem cell pool. In this manner, metabolic interventions that alter stem cell survival or differentiation in response to stress could result in long-term adverse consequences for organ function.

Integrated stress response

Various cellular stresses converge on common response pathways. One such pathway is the integrated stress response (ISR), which responds to conditions such as impaired proteostasis, nutrient deprivation, mitochondrial dysfunction, viral infection, and oxidative stress.[133] ISR activation is initiated by specialized kinases that act as sensors for diverse cellular pathways including metabolism by phosphorylating the alpha subunit of eukaryotic factor 2 (eIF2α) to either promote cell survival or facilitate cell death in the event of unmitigable cellular stress. Recent studies have identified specific pathways that relay mitochondrial stress via the ISR to trigger adaptive cell responses.[134,135]

While the ISR has been investigated extensively in cancer and neurodegenerative disease, less is known about ISR function in stem cells,[133] though numerous studies have demonstrated differences in ISR activation across stem cell states. In the hematopoietic system, increased basal ISR activity in HSCs leads to apoptosis downstream of PERK activation and ATF4 translation.[127] HSCs exhibited higher expression of an ATF4 reporter than their downstream committed progenitors, and ATF4^{hi} cells had increased engraftment in transplantation experiments.[136] Similar results were seen in E4.5 pre-implantation epiblast cells,[137] in naive PSCs,[138] and in MuSCs,[86] which all exhibit high levels of *p*-eIF2α that decrease with differentiation. ISR activation not only tracks with stemness but can play a causative role. Replacement of the phosphorylated serine residue in *p*-eIF2α by a non-phosphorylatable alanine residue impaired PSC proliferation and pluripotency upon removal of factors that maintain the naive state.[137] Similarly, inhibition of the eIF2α phosphatase GADD34/PP1 promotes both PSC[137] and MuSC self-renewal.[86]

Conversely, in ISCs *p*-eIF2α is inversely correlated with stem cells markers and the ISR is induced downstream of ER stress upon differentiation to transit-amplifying cells.[132] Additionally, the ISR is transiently upregulated in lung development but can also be pathologically activated by loss of mitochondrial complex I,[139] suggesting that metabolic pathways may act to constrain the ISR in homeostatic conditions. Experiments in myoblasts and differentiated myotubes *in vitro* further demonstrated that the metabolic stressors that lead to ISR activation vary according to cell state.[140] Therefore, the pathways that trigger ISR activation in different stem cells are likely highly context dependent.

Emerging evidence suggests that the ISR might maintain stem cell state independent of cellular stressors. While most studies of the ISR have focused on transcription factors such as ATF4 or CHOP that mediate cell survival, ISR activation can lead to selective translation of additional proteins with uORFs that have known functions in stem cells.[86,137] These include the pluripotency transcription factors NANOG and MYC that demonstrate increased levels in PSCs following eIF2α phosphatase inhibition.[138] Collectively, these studies suggest that ISR activation may be rheostatic, where a basal level of ISR signaling maintains stemness but in the setting of cellular stress is maladaptive. Future exploration is critical to better understand how stem cell metabolism controls ISR engagement to selectively activate these disparate pathways and dictate cellular outcomes.

Antioxidant and detoxification pathways

Stem cells have additional strategies to directly mitigate endogenous toxic metabolites and preserve the stem cell state. Perhaps the most studied of these metabolites are reactive oxygen species (ROS), which are oxygen-containing free radicals that in excess can be cytotoxic. ROS are a direct byproduct of cell metabolism, and the role of ROS in stem cell maintenance and differentiation has been extensively reviewed.[141–143] Notably, alterations in ROS levels cause both failure of stem cell self-renewal and impaired differentiation, suggesting that tight regulation of ROS is critical for stem cell function.

Because ROS can be genotoxic, many stem cell populations moderate its production to ensure genetic integrity in naive progenitors. An example of this can be seen in PSCs, where differentiation is accompanied by an increase in oxidative metabolism and ROS levels.[144] Similar patterns of increasing oxidation with commitment are seen in HSCs, where ROS levels increase with progenitor commitment and cells with low ROS have the highest potential for self-renewal in serial transplantation assays.[145] Mitochondrial ETC complexes I and III are a major source of ROS production in mammalian cells; therefore, decreased electron deposition into the ETC is one method of controlling cellular ROS levels. Not surprisingly, increased ROS levels are correlated with increased respiration and stem cell dysfunction in a variety of genetic models, and antioxidant treatment rescues stem cell defects seen in settings of altered respiration.[47,100,101,146]

In contrast to HSCs where increasing ROS correlates with reduced self-renewal, increasing ROS in NSCs correlates with stem cell proliferation *in vivo*, and supplementation of low levels of ROS boosts the capacity of NSCs for *in vitro* neurosphere formation.[147] Similarly, treatment with the mitochondria-targeted antioxidant ubiquinone was toxic to both PSCs and NSCs.[146] These results fit a growing theme that while high ROS levels can be toxic to stem cells, moderate levels of ROS are necessary for proliferation and differentiation. In support of this model, antioxidant treatment blocks differentiation in multiple stem cell types.[148,149]

Endogenous antioxidant systems also play critical roles in stem cell biology. One pathway that has received considerable attention is the FOXO transcription factors, which have evolutionary conserved functions in regulating ROS levels.[150] Conditional deletion of one or more FOXO transcription factors is pathogenic in multiple stem cell types including PSCs,[151]

HSCs,[152–154] NSCs,[155–157] and MuSCs.[158] Across these studies, loss of FOXO correlates with increased ROS levels and defective stem cell function. However, the mechanisms by which FOXO regulates ROS in stem cells remains unclear. FOXO transcription factors can transcriptionally regulate antioxidant enzymes and DNA damage response pathways.[150] Other studies in HSCs have demonstrated that loss of FOXO3 can affect mitochondrial metabolism, either through autophagy[159] or through regulation of mitochondria.[160] It is reasonable to hypothesize that FOXO and other antioxidant pathways have specific roles in different stem cell types, and their dysregulation plays a role in stem cell aging or disease. To the extent that both antioxidants and toxic metabolites are the products of metabolic reactions, the specific metabolic preferences of stem cells almost certainly affect both the propensity for ROS generation and detoxification, providing an additional link between metabolic networks and stem cell fate.

HALLMARK 4: METABOLIC CONTROL OF STEM CELL FATE VIA EPIGENETIC MODIFICATION

In addition to meeting the demands of proliferation and resisting cellular stressors, stem cells must retain the capacity for differentiation. Stem cell differentiation is principally controlled at the level of transcription, where cell state transitions are coordinated by the activity of chromatin remodelers and the binding of lineage-specific transcription factors. Notably, metabolites are the chemical precursors for post-translational modifications of histones and nucleic acids. This link between metabolites and chromatin has led to the hypothesis that metabolic fluctuations can alter the distribution of chromatin and DNA modifications. In this scenario, metabolites serve not just as substrates for growth or viability, but also as signals that influence gene expression programs to alter cell fate.[141] While this hypothesis has been interrogated in a rapidly growing body of literature, particularly in the field of cancer metabolism,[25,161] fewer studies have examined metabolic and epigenetic crosstalk in stem cells.

Histone acetylation

Acetylation is a ubiquitous post-translation modification that decorates histones and is associated with increased chromatin accessibility and active gene transcription.[162] Histone acetylation is catalyzed by histone acetyltransferases (HAT) that use the metabolite acetyl-CoA as an acetyl-donor.[163] Acetyl-CoA is generated by numerous pathways in the cell, including mitochondrial oxidation of glucose, amino acids, and fatty acids (Figure 5A). This pool of acetyl-CoA cannot directly exit the mitochondria; rather, acetyl-CoA produces citrate that when exported from mitochondria is cleaved by ATP-citrate lyase (ACL) in the cytosol, yielding acetyl-CoA for acetylation and *de novo* lipid synthesis.[164] Cytosolic acetyl-CoA can also be directly generated from acetate via the action of a cytosolic acetyl-CoA synthetase (ACSS2). Cells can still generate some cytosolic acetyl-CoA even in the complete absence of ACL and ACSS2, likely via mitochondrial carnitine shuttles and/or peroxisomal metabolism,[165,166] underscoring the deep redundancy of metabolic networks in mammalian cells.

The source of cytosolic acetyl-CoA has been shown to vary with cellular state. In PSCs cultured in the naive ground state of pluripotency, glucose-derived citrate is largely retained in the mitochondria and loss of ACL has modest effects on overall levels of histone acetylation.[40] Upon exit from the naive ground state, mitochondrial-derived citrate becomes a major source of cytosolic acetyl-CoA and loss of ACL consequently impairs H3K27 and H4K16 acetylation. Providing exogenous acetate to boost acetyl-CoA production promoted histone acetylation and delayed differentiation in both human and mouse PSCs.[40,44] In mouse PSCs, threonine also contributes to acetyl-CoA pools due to unique expression of the enzyme threonine dehydrogenase.[66] ACL-derived acetyl-CoA is likewise required for adult stem cells. In adipocyte differentiation *in vitro*, silencing of ACL leads to lower levels of pan-histone acetylation, altered gene expression, and diminished lipid accumulation.[164] Similar effects were seen in myoblasts, where ACL knockdown resulted in reduced histone acetylation at the *MYOD* promoter and impaired differentiation to myotubes.[167] Collectively these studies indicate that changing availability of cytosolic acetyl-CoA can link metabolic pathways with gene expression programs in a variety of stem cell populations.

Histone acetylation turns over rapidly,[168] but deacetylation by a subset of HDACs may also be subject to metabolic regulation. One class of HDACs is the sirtuins, a family of NAD^+-dependent deacylases that cleave NAD^+ as they remove acyl groups including acetyl moities from various proteins.[169] While several sirtuins have demonstrated roles in stem cell function,[169] it is less clear whether changes in cellular NAD^+ levels contribute to cell fate control by modulating sirtuin activity. More conventional HDACs can be inhibited by metabolites such as ketone bodies, providing an additional mechanism beyond acetyl-CoA availability for metabolism to alter histone acetylation. In ISCs, expression of the enzyme HMGCS2 controls levels of β-OHB which acts as an HDAC inhibitor to reinforce ISC self-renewal programs,[170] raising the possibility that changes in ketone body metabolism may affect a variety of stem cell populations. Indeed, many intracellular metabolites have the potential to impact histone acetylation: HATs can use a variety of acyl-CoA substrates, often in proportion to their cellular abundance.[171] Although we lack a clear picture of how cells control levels of various acyl-CoA species and how competition for HATs affects gene expression outcomes, the ability of multiple metabolic intermediates to modulate histone acylation provides another potential avenue for cells to link nutrient abundance and metabolic networks with cell fate control.

Histone and DNA methylation

Histones can additionally be modified by methylation of specific lysine or arginine residues, each of which has been shown to have unique effects on transcriptional activity.[172] Methylation is linked to nutrient availability through levels of S-adenosylmethionine (SAM), the universal methyl donor for enzymes that transfer methyl groups to target substrates[173] (Figure 5B). In most cells, methionine is the major source of SAM, but serine and glycine metabolism can also generate one-carbon units that fuel SAM regeneration from S-adenosylhomocysteine (SAH).[174] Fluctuations in each of these pathways are proposed to alter SAM levels and downstream histone or DNA methylation in diverse cell types, including PSCs that must maintain a globally hypomethylated landscape prior

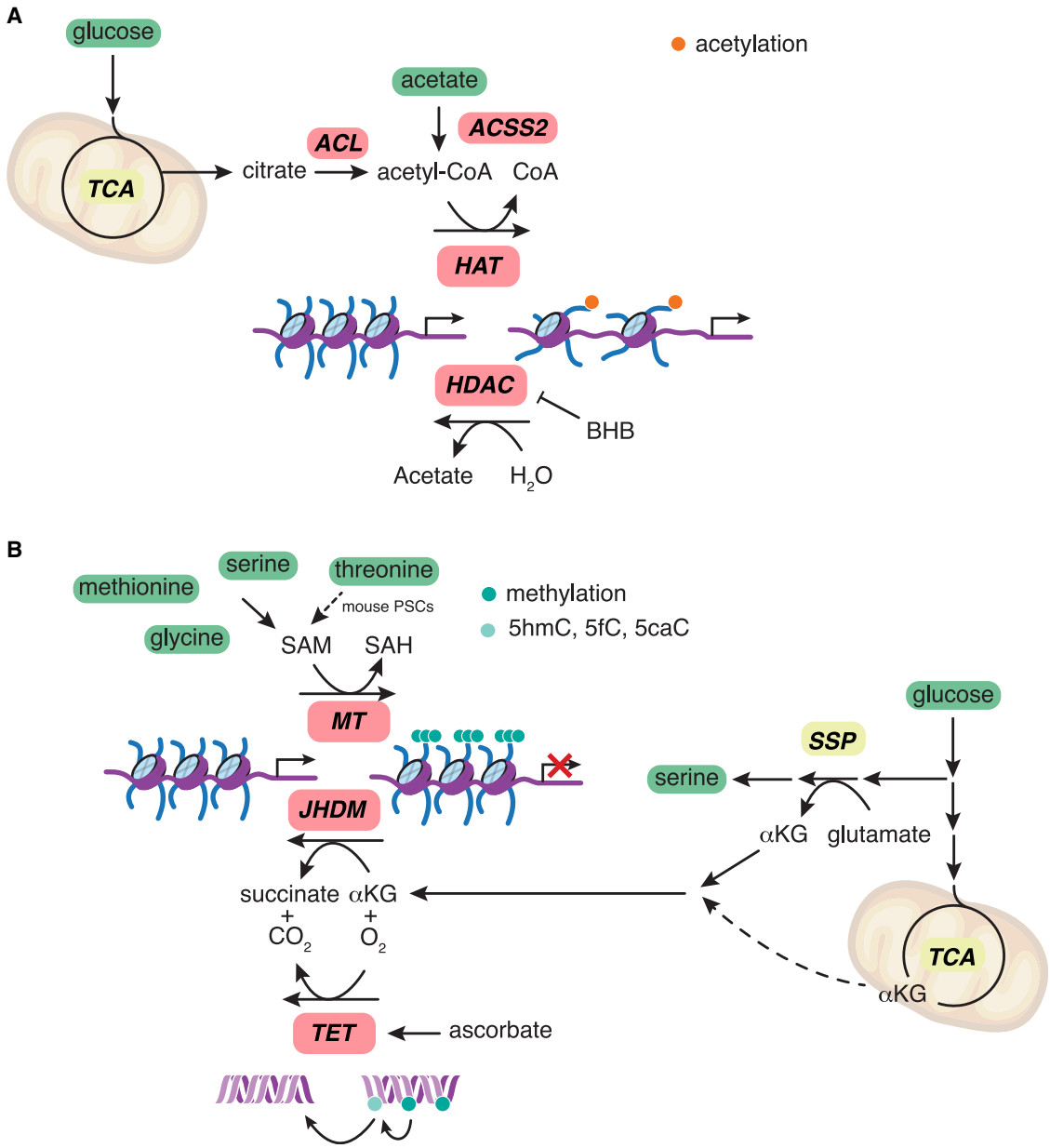

Figure 5. Metabolic-chromatin crosstalk in stem cells

Metabolites form the chemical modifications that decorate chromatin and can regulate the activity of many chromatin-modifying enzymes to regulate stem cell fate.

(A) Cytosolic acetyl-CoA is the acetyl-donor for histone acetyltransferases and is largely generated either from mitochondrially derived citrate by ACL or from acetate by ACSS2. Ketone bodies such as BHB have been shown to inhibit histone deacylases.

(B) Multiple amino acids can contribute to cellular pools of SAM, the methyl donor for histone and DNA methylation by methyltransferases. Demethylation is carried out by the family of α-KG-dependent-dioxygenases. Cellular levels of α-KG influence dioxygenase activity through its role as a co-substrate for demethylation reactions. Ascorbate is a cofactor that promotes the activity of the TET enzymes that iteratively oxidize methylated cytosines in DNA generate 5-hydroxymethylcytosine (5hmC) or its oxidized derivatives (5-formylcytosine or 5-carboxycytosine) that are intermediates in DNA demethylation.

to lineage commitment. Complete methionine deprivation in human PSCs rapidly decreases SAM pools and leads to cell death along with a decrease in histone 3 lysine 4 trimethylation (H3K4me3).[69] Further studies have shown that mouse PSCs catabolize threonine to acetyl-CoA and glycine that can contribute to SAM synthesis, and threonine withdrawal in these cells lowers H3K4me3. However, measurements of SAM in mouse ESCs using a ratiometric RNA-based sensor demonstrated no change in SAM levels upon threonine withdrawal, in contrast to methionine withdrawal which led to rapid loss of sensor signal.[175] These experiments demonstrate that SAM levels must be tightly regulated to ensure proper PSC function, but suggest further studies are necessary to conclude how SAM levels are controlled endogenously in different systems.

Indeed, increased SAM consumption—not just SAM production—may be a major determinant of SAM levels: in naive PSCs, high expression of the SAM-consuming enzyme nicotinamide *N*-methyltransferase can limit SAM levels, thus decreasing tri-methylation at histone 3 lysine 27 (H3K27me3).[176]

Histone and DNA methylation can also be regulated by levels of the TCA cycle metabolite alpha-ketoglutarate (α-KG). α-KG is a co-substrate for the large family of α-KG-dependent dioxygenases, including the Jumonji-domain containing histone demethylases and the ten-eleven-translocation (TET) enzymes that facilitate the hydroxylation and demethylation of many methylated histones and DNA methylcytosine. In both PSCs and a variety of other stem and cancer cell types, α-KG has been proposed to control cell fate programs via these α-KG-dependent demethylases that remove repressive DNA or histone methylation.[177] However, manipulating α-KG has diverse effects in different stem cell populations, indicating that metabolic regulation of chromatin demethylation is likely extremely context dependent.

Mouse PSCs cultured under conditions that promote the naive state of pluripotency exhibit increased intracellular levels of α-KG.[38,178] In these cells, the addition of cell-permeable α-KG supports PSC self-renewal,[38] while conversely in primed human PSCs α-KG supplementation leads to differentiation accompanied by DNA and histone demethylation.[179] These disparate outcomes may be due to the very different chromatin landscapes of naive and primed PSCs: naive ESCs are globally hypomethylated, and differentiation requires methylation for silencing pluripotency transcriptional networks.[180,181] In contrast, primed cells with somatic levels of methylation likely require demethylation to induce lineage-specific genes.

Consistent with this model, α-KG promotes differentiation in several adult stem cell populations. In epidermal stem cells, *de novo* serine synthesis produces α-KG that drives loss of H3K27me3 and promotes epidermal differentiation.[177] Exogenous α-KG supplementation likewise induced differentiation in HSCs[99] and intestinal organoids.[182] Across these studies, a myriad of α-KG-utilizing enzymes have been implicated as the drivers of α-KG-dependent effects on cell fate. Likely, the role of α-KG will depend on the complement of dioxygenases expressed in a given cell type and the particular chromatin barriers to a particular fate change.

Additional evidence for metabolic regulation of chromatin methylation comes from the function of ascorbate, a cofactor that supports activity of α-KG-dependent dioxygenases. Ascorbate addition in PSCs promotes TET1 activity and leads to an increase in DNA demethylation.[183] In the hematopoietic system, high levels of ascorbate are maintained in HSCs and decrease with differentiation to progenitors. Mice unable to synthesize ascorbate had increased HSC frequency with reduced levels of 5hmC, suggestive of reduced TET activity.[184] By increasing levels of either the co-substrate (α-KG) or enzyme activity (ascorbate), stem cells might regulate chromatin methylation states in response to intracellular metabolic rewiring.

Furthermore, mutations in metabolic enzymes that oppose these pathways have been shown to dysregulate stem cell function. Recurrent mutations in isocitrate dehydrogenase 1 and 2 (IDH) that lead to the production of 2-hydroxyglutarate (2-HG)—an oncometabolite that is a competitive inhibitor of α-KG-dependent dioxygenases—have been identified in multiple cancer types.[177] Mutant IDH is sufficient to impair differentiation in multiple tissue lineages, consistent with the notion that 2-HG and subsequent dioxygenase repression promote aberrant stem cell self-renewal.[185–189] Therefore, genetic alterations that lead to tumorigenesis might co-opt metabolic-epigenetic regulation that underlies normal stem cell function.

Like 2-HG, the TCA cycle intermediates succinate and fumarate can also serve as competitive inhibitors of α-KG-dependent dioxygenases, and both metabolites accumulate to pathological levels in tumors with mutations in genes encoding subunits of succinate dehydrogenase (SDH) or fumarate hydratase (FH), respectively.[177] Emerging evidence indicates that SDH and FH are required for normal stem cell function. Inducible deletion of SDH subunit *Sdhb* led to death of HSCs, although whether this was because of succinate accumulation specifically, or a blockade in the TCA cycle more generally, remains to be determined.[190] Like SDH loss, FH loss impaired normal hematopoiesis, but, notably, hematopoiesis was restored by a cytosolic FH variant that normalized fumarate levels without rescuing mitochondrial substrate oxidation.[191] These results suggest that fumarate accumulation, rather than mitochondrial TCA cycle failure, impaired normal hematopoiesis. Major open questions are how perturbations in the TCA cycle affect stem cell function and whether TCA cycle-derived metabolites affect cell fate control through α-KG-dependent dioxygenases or through other mechanisms (for example, fumarate is an electrophile that can directly modify reactive cysteines).

The above studies clearly illustrate that fluctuations in metabolic pathways can alter the levels of chromatin and DNA modifications and that these alterations perturb downstream stem cell function. It will be important for future studies to assess whether normal physiologic shifts in metabolite concentrations are sufficient to modulate the function of enzymes controlling the chromatin landscape.

HALLMARK 5: METABOLIC REQUIREMENTS OF THE STEM CELL NICHE

A critical component of stem cell biology is the endogenous niche. Biophysical or paracrine signals from other stem cells or niche components regulate stem cell function and are often dysregulated in pathologies and aging.[192] An emerging question is whether characteristics of the niche support stem cell metabolism. Reciprocally, specific metabolic adaptations might enable stem cell survival in specialized niche environments.

Proximity to vasculature and lymphatics

A key feature of the stem cell niche is that stem cells are spatially located in fixed proximity to defined anatomical structures. Physical distance from structures such as vasculature can create regions of limited oxygen concentration or nutrient availability that pose metabolic challenges for resident stem cells. The best evidence for this is seen in HSCs, which reside in endosteal zones of the bone marrow adjacent to sinusoidal blood vessels.[193] These regions are among the most hypoxic in the bone marrow,[194,195] suggesting that HSC function must be optimized for low oxygen levels. The role of hypoxia in the HSC niche is independent of the hypoxia-inducible transcription

factors HIF1α and HIF2α, which are not required for HSC self-renewal.[196,197] Hypoxia may play a role in limiting ROS levels in HSCs, as transient exposure of HSCs to atmospheric oxygen increases ROS and impairs HSCs.[198]

Like HSCs, NSCs are perivascular but are thought to reside in regions of the brain with relatively higher oxygen tension.[199,200] To date, the role of oxygen levels in NSCs or other tissue stem cells remains unclear. Studies exploring the HFSC and ISC niche have identified lymphatics as critical niche components that directly associate with stem cells.[201–204] While these studies have principally focused on signaling between stem cells and lymphatic endothelial cells, it is possible that lymphatic drainage may support metabolism as well by removing toxic metabolites or waste products.

Nutrient acquisition strategies

Whereas *in vitro* culture systems provide nutrients at levels often in vast excess of cellular requirements, nutrient limitation might represent a barrier to stem cell function in their endogenous niche. Testing this hypothesis is limited by technical challenges in measuring metabolite concentrations in the interstitial fluid of different compartments. Suggestively, studies measuring amino acid concentrations in the bone marrow revealed a distinct amino acid profile from the peripheral blood and identified valine as selectively required for HSC homeostasis.[205] However, the presence of an amino acid in the niche does not guarantee its uptake. For example, aspartate availability is limiting for HSCs during hematopoietic regeneration, and overexpression of the glutamate/aspartate transporter was sufficient to increase aspartate levels and HSC function.[206] Therefore, nutrient availability must be coupled with acquisition strategies to ensure optimal stem cell function.

PSCs are a transient state captured *in vitro* with no direct equivalent during *in vivo* development. Nevertheless, they retain several metabolic aspects of blastocysts that reside in the maternal reproductive tract. While glucose and soluble amino acids are an important nutrient source for mammalian cell proliferation, the maternal reproductive tract is relatively nutrient poor.[207] In most mammalian cells, amino acids are acquired by a diverse set of plasma membrane transporters that facilitate entry of these nutrients to fuel protein synthesis and other cellular functions.[208] In contrast, naive PSCs have reduced levels of these plasma membrane amino acid transporters and can proliferate and gain biomass in the absence of most essential amino acids.[111] Mechanistically, amino acid independence is enabled by high levels of macropinocytosis, whereby extracellular protein is taken up and digested in the lysosome to release amino acids required for growth.[111]

The ability of naive PSCs to grow without essential amino acids mimics early studies demonstrating that embryos can develop to the blastocyst stage in the complete absence of soluble amino acids.[209,210] Indeed, all cells of the pre-implantation blastocyst engage in macropinocytosis,[111] suggesting a conservation of nutrient acquisition programs between PSC culture models and embryonic development. That these metabolic strategies are hard-wired into the cells of the pre-implantation epiblast irrespective of other cell types or 3D architecture verifies their utility to uncover metabolic programs with direct relevance to embryogenesis.

Some evidence suggests phagocytosis plays a role in other stem cell populations. Basal epithelial cells and HFSCs activate phagocytosis in the normal hair cycle to clear dying apoptotic cells.[211,212] While these studies have focused on how phagocytosis maintains tissue architecture, uptake and degradation of extracellular macromolecules could support cell metabolism. Stem cells likewise interact with other cell types within the niche, and it remains to be seen whether niche cells specifically support stem cell metabolism. In the intestine, monocarboxylate production by intestinal niche cells enhances oxidative phosphorylation by ISCs.[213] Other stem cell-niche interactions might have similar unexplored metabolic benefits. Finally, dietary interventions can alter nutrient availability in the stem cell niche and regulate stem cell metabolism to impact cell fate.[214]

FUTURE DIRECTIONS

Ongoing studies continue to explore the relationship between metabolism and stem cell function. Despite the exponential progress that has been made since the first isolation of adult and embryonic stem cells over 50 years ago, our view of stem cell metabolism is still limited. In the field of stem cell biology at large, the popularization of single-cell sequencing technology has increasingly illuminated the diversity of distinct cell states within stem cell populations.[215] This increasing transcriptomic resolution continues to redefine our definition of what constitutes the pluripotent or multipotent stem cell pool. An important corollary to these sequencing studies is the development of protocols to measure metabolites in increasingly small numbers of cells that often must be isolated from their tissue of origin.[19,20] The ability to correlate metabolic and transcriptomic profiles will be crucial for defining how metabolism can control cell fate across different stem cell types. In parallel with continued exploration of phenotypic heterogeneity within stem cell populations, increased attention must be paid to the study of stem cell metabolism in their endogenous context.[7] Technological advances in techniques such as spatial metabolomics[216] will deepen our understanding of how metabolite levels relate to positioning within 3D tissue architecture. These tools have already been applied to both normal tissues and malignancies[217–219] and will be of use to explore how stem cell metabolism relates to position in the niche and proximity to other niche structures.

To date, most of our understanding of metabolic control of stem cell fate relies on experiments where metabolites are supplemented either in the diet or in culture media, or upon deletion of metabolic genes in the stem cell compartment. These approaches reflect the ways metabolic enzymes can be mutated in cancer or inborn errors of metabolism, leading to the buildup of supraphysiologic metabolite levels. Expanding the use of physiologic media[220] that better recapitulates the nutrient composition of different stem cell environments will improve the physiologic relevance of *in vitro* stem cell models. Such approaches will allow us to understand whether physiological shifts in metabolites alter stem cell function and how stem cells preserve their longevity and identity in the face of a changing microenvironment.

Ultimately, many questions will require the ability to perform isotope tracing in stem cells *in vivo*. Recent studies have pioneered new approaches to perform isotope labeling in the

placenta and embryo during midgestation[221] and mid-to-late gestation[222] in mice, opening the possibility of monitoring embryonic metabolism during normal intrauterine development and in disease. Similarly, new stem-cell-derived models of mammalian embryogenesis continue to be refined.[223] These tools have already led to studies probing the relationship between metabolism and development[224–226] and will undoubtedly help bridge the gap between *in vitro* and *in vivo* studies of developmental metabolism.

While the increased ability to measure metabolism in stem cells will be integral toward defining what stem cell metabolism *is*, metabolite levels alone are insufficient to fully understand how metabolic pathways influence stem cell biology. Rather, there is a pressing need for studies that ask *how* metabolism supports stem cell function. In addition to advancing tools that measure increasing complexity within the stem cell population, it remains important to utilize models that conversely take a reductionist approach. During normal stem cell homeostasis, cell proliferation is tightly linked to cell fate and location within the niche. Consequently, interventions that alter any of these individual components will likely affect others in turn. In the case of metabolic perturbations, this challenges our ability to link specific metabolic fluctuations to any individual stem cell attribute. It is therefore critical to develop systems to isolate individual stem cell attributes and allow careful interrogation of metabolic manipulations, while in parallel asking questions that seek to ascribe metabolic regulation to specific stem cell outputs. Subsequently, metabolic principles derived from comparatively simpler experiments can be extrapolated to stem cells in living organisms. Together, these approaches are likely to elucidate fundamental principles of stem cell metabolism and provide opportunities to improve stem cell function in disease, tissue regeneration, and therapeutic interventions.

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DECLARATION OF INTERESTS

B.T.J. and L.W.S.F. are listed as inventors on patent application(s) related to metabolism and cell fate control in embryonic stem cells.

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